

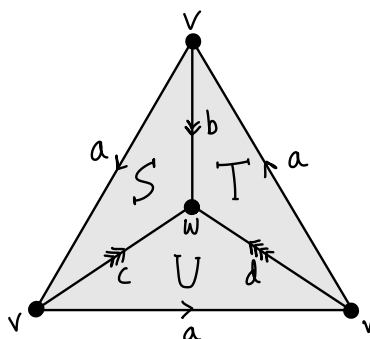
PhD Topology Exam

May 2025

Be neat, do not write too small, and leave margins.

For the first five problems, show all of your work and support all statements. Use a separate sheet of paper for each problem. (Each problem 10 points.)

1. Show that the product of two connected spaces is connected.
2. Prove that every metrizable space (X, d) is normal.
3. Let X be the Δ -complex drawn below, which has two 0-simplices v and w , four 1-simplices a, b, c, d , and three 2-simplices S, T, U . Compute the simplicial homology $H_i(X)$ for $i = 0, 1, 2$.



4. (a) For $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ a short exact sequence of finitely generated abelian groups, prove $\text{rank}(B) = \text{rank}(A) + \text{rank}(C)$.
 (b) For $\dots \rightarrow C_n \xrightarrow{d_n} C_{n-1} \rightarrow \dots$ a chain complex of finitely generated abelian groups, prove

$$\text{rank}(C_n) = \text{rank}(\ker d_n) + \text{rank}(\text{im } d_n) \quad \text{and} \quad \text{rank}(\ker d_n) = \text{rank}(\text{im } d_{n+1}) + \text{rank}\left(\frac{\ker d_n}{\text{im } d_{n+1}}\right).$$

 (c) Prove a finite CW complex X with c_n cells in dimension n satisfies

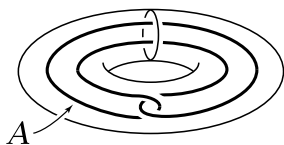
$$\sum_n (-1)^n c_n = \sum_n (-1)^n \text{rank}(H_n(X)).$$

5. (a) Let $n > m$. Show there are no maps from $\mathbb{RP}^n$ to $\mathbb{RP}^m$ inducing a nontrivial map $H^1(\mathbb{RP}^m; \mathbb{Z}_2) \rightarrow H^1(\mathbb{RP}^n; \mathbb{Z}_2)$.
 (b) Prove the Borsuk–Ulam theorem as follows. Suppose on the contrary that $f: S^n \rightarrow \mathbb{R}^n$ satisfies $f(x) \neq f(-x)$ for all x . Define $g: S^n \rightarrow S^{n-1}$ by $g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$, so $g(-x) = -g(x)$ and g induces a map $\mathbb{RP}^n \rightarrow \mathbb{RP}^{n-1}$. Show that (a) applies to this map.

The remaining problems are on the following page.

Answer the following ten problems with complete definitions, complete statements, an example, or a short proof. (Each problem 5 points.)

6. State the Urysohn Lemma.
7. Give an example of a space that is path-connected but not locally path-connected
8. Use the Seifert van-Kampen theorem to derive the fundamental group of M_g , the orientable surface of genus g , for $g \geq 1$.
9. Show that for any connected closed orientable n -manifold M there is a degree 1 map $M \rightarrow S^n$.
10. Show that every map $f: \mathbb{CP}^n \rightarrow \mathbb{CP}^n$ has a fixed point if n is even.
11. Show that $S^1 \vee S^1$ admits a covering space whose fundamental group is nontrivial and abelian.
12. Let $S^1 \times D^2$ be the solid torus, and let $A \subseteq S^1 \times D^2$ be the circle shown in the figure. Prove that no retraction $r: S^1 \times D^2 \rightarrow A$ exists.



13. Show that if $f: X \rightarrow Y$ is a continuous bijection with X compact and Y Hausdorff, then f is a homeomorphism.
14. State the Baire Category Theorem, and use it to prove that $[0, 1]$ is uncountable.
15. Does there exist a closed orientable 14-dimensional manifold M with $H^7(M; \mathbb{Z}) \cong \mathbb{Z}$?